Characterising split opfibrations using lenses

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Wednesday 4 December 2019

63rd Annual Meeting of the Australian Mathematical Society

CoACT, Macquarie University

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- **Problem**: these definitions appear to be quite different.
- **Solution**: characterise split opfibrations using delta lenses. Split opfibrations = (functor + lifting) + universal property

Background

Discrete opfibrations

Definition

A functor $G\colon \mathbf{S} \to \mathbf{V}$ is a discrete opfibration if for all objects S in $\mathbf{S}$ and morphisms $\alpha\colon GS \to V$ in $\mathbf{V}$, there exists a unique morphism $\widehat{\alpha}\colon S \to S'$ such that $G\widehat{\alpha} = \alpha$.

$$\begin{tikzcd} \mathbf{S} \arrow[d, "G"'] & S \arrow[r, dashed, "\widehat{\alpha}"] & S' \\ \mathbf{V} & GS \arrow[r, "\alpha"] \arrow[u, dotted, no head] & V \arrow[u, dotted, no head] \end{tikzcd}$$

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$$\begin{tikzcd} \mathbf{S} \arrow[d, "G"'] \\ \mathbf{V} \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, dashed, "\widehat{\alpha}"] & S' \\ \vdots & \vdots \\ GS \arrow[r, "\alpha"] & V \end{tikzcd}$$

Every functor $F\colon \mathbf{V} \to \mathbf{Set}$ yields a discrete opfibration $\int F \to \mathbf{V}$ via Grothendieck construction, and vice versa.

Opcartesian morphisms for a functor $G\colon \mathbf{S} \to \mathbf{V}$

Definition

A morphism $u\colon S \to S'$ in $\mathbf{S}$ is opcartesian if for all morphisms $f\colon S \to S''$ in $\mathbf{S}$ and $\beta\colon GS' \to GS''$ in $\mathbf{V}$ such that $\beta \circ Gu = Gf$, there exists a unique morphism $\widehat{\beta}\colon S' \to S''$ such that $G\widehat{\beta} = \beta$ and $\widehat{\beta} \circ u = f$.

$$
\begin{tikzcd}
\mathbf{S} \arrow[d, "G"'] & S \arrow[r, "u"'] \arrow[rr, bend left, "f"] & S' \arrow[r, dashed, "\widehat{\beta}"'] & S'' \\
\mathbf{V} & GS \arrow[r, "Gu"] \arrow[rrr, bend right, "Gf"'] & GS' \arrow[r, "\beta"] & GS''
\end{tikzcd}
$$

Thus opcartesian morphisms satisfy a *universal property*.

Split opfibrations

Definition

A split opfibration is a pair (G, k) consisting of a functor $G\colon \mathbf{S} \to \mathbf{V}$ together with a function k, called a *splitting*,

$$(S, \alpha\colon GS \to V) \quad \longmapsto \quad k(S, \alpha)\colon S \to p(S, \alpha)$$

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Every functor $F\colon \mathbf{V} \to \mathbf{Cat}$ yields a split opfibration $\int F \to \mathbf{V}$ via Grothendieck construction, and vice versa.

Characterising split opfibrations using lenses and décalage

Delta lenses

Definition (Diskin, Xiong, Czarnecki, 2011)

A delta lens is a pair (G, k) consisting of a functor $G\colon \mathbf{S} \to \mathbf{V}$ together with a function k, called a *lifting* (or *put*),

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A split opfibration is a delta lens where each $k(S, \alpha)$ is opcartesian.

Characterising delta lenses using functors

Lemma

Every delta lens $(G, k)\colon \mathbf{S} \to \mathbf{V}$ *induces a category* $\mathbf{\Lambda}$ *with the same objects as* $\mathbf{S}$ *and with morphisms given by formal pairs:*

$$S \xrightarrow{(S,\alpha)} p(S, \alpha)$$

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Proposition

Every delta lens may be represented as a diagram in **Cat** *given by,*

$$\begin{tikzcd} & \mathbf{\Lambda} \arrow[dl, "K"'] \arrow[dr, "\overline{G}"] & \\ \mathbf{S} \arrow[rr, "G"'] & & \mathbf{V} \end{tikzcd}$$

where K *is identity-on-objects and* $\overline{G}$ *is a discrete opfibration.*

Décalage

Definition

There is a comonad $\mathrm{Dec}_{\mathrm{r}}$ on **Cat** called the right décalage which assigns to each category $\mathbf{S}$ the sum of its slice categories:

$$\mathrm{Dec}_{\mathrm{r}}(\mathbf{S}) = \sum_{X \in \mathbf{S}} \mathbf{S}/X$$

The counit components are *discrete fibrations* given by:

$$\begin{tikzcd} S \arrow[rr, "f"] \arrow[dr, "gf"'] & & S' \arrow[dl, "g"] \\ & X & \end{tikzcd} \quad \longmapsto \quad S \xrightarrow{f} S'$$

Main Theorem

Theorem

A delta lens $(G, k)\colon \mathbf{S} \to \mathbf{V}$ *is a split opfibration if and only if the composite functor* F *shown below is a discrete opfibration.*

$$\begin{tikzcd}
\bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] \arrow[rr, bend left, "F"] & \mathrm{Dec_r}(\mathbf{S}) \arrow[r] \arrow[d] & \mathrm{Dec_r}(\mathbf{V}) \arrow[d] \\
\mathbf{\Lambda} \arrow[r, "K"] \arrow[rr, bend right, "\overline{G}"'] & \mathbf{S} \arrow[r, "G"] & \mathbf{V}
\end{tikzcd}$$

Split opfibration = delta lens + universal property

Characterising split opfibrations using lenses and double categories

Split opfibrations as algebras for a monad

Let $\mathbf{V}$ be a category. There is a monad M on the slice category $\mathbf{Cat} \,/\, \mathbf{V}$ whose action on objects is given by,

$$G \colon \mathbf{S} \to \mathbf{V} \quad \longmapsto \quad \Pi_{\mathbf{V}} \colon G \downarrow \mathbf{V} \to \mathbf{V}$$

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Unpacking: split opfibration is a pair (G, P) consisting of functors $G\colon \mathbf{S} \to \mathbf{V}$ and $P\colon G \downarrow \mathbf{V} \to \mathbf{S}$ such that $GP = \Pi_{\mathbf{V}}$.

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Split opfibration = functor + structure map

Generalises to any 2-category with pullbacks and comma objects.

Does this definition make sense?

Consider a split opfibration given by the pair (G, P), where the functor $P\colon G \downarrow \mathbf{V} \to \mathbf{S}$ has action on morphisms given by:

$$\begin{tikzcd} GS \arrow[r, "\alpha"] \arrow[d, "G(f)"'] & V \arrow[d, "\beta"] \\ GS' \arrow[r, "\alpha'"'] & V' \end{tikzcd} \quad \longmapsto \quad \begin{tikzcd} P(S, \alpha) \arrow[d, "P\langle f, \beta\rangle"] \\ P(S', \alpha') \end{tikzcd}$$

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Proposition

Every split opfibration (G, P) yields a split opfibration of the form (G, k) where $k(S, \alpha) = P\langle 1_S, \alpha\rangle$ as depicted below:

$$\begin{tikzcd} GS \arrow[r, "1_{GS}"] \arrow[d, "G(1_S)"'] & GS \arrow[d, "\alpha"] \\ GS \arrow[r, "\alpha"'] & V \end{tikzcd} \quad \longmapsto \quad \begin{tikzcd} S \arrow[d, "P\langle 1_S, \alpha\rangle"] \\ P(S, \alpha) \end{tikzcd}$$

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Lemma

Every split opfibration $(G, P)\colon \mathbf{S} \to \mathbf{V}$ *induces a double category* $\mathbb{A}$ *with domain and codomain maps given by* $\Pi_{\mathbf{S}}, P\colon G \downarrow \mathbf{V} \rightrightarrows \mathbf{S}$.

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Every split opfibration may be represented as a diagram in **Dbl** *given by,*

$$\begin{tikzcd} & \mathbb{A} \arrow[dl, "\mathcal{K}"'] \arrow[dr, "\overline{\mathcal{G}}"] & \\ \mathrm{Sq}(\mathbf{S}) \arrow[rr, "\mathrm{Sq}(G)"'] & & \mathrm{Sq}(\mathbf{V}) \end{tikzcd}$$

where $\mathcal{K}$ *is identity-on-objects and* $\overline{\mathcal{G}}$ *is a discrete opfibration.*

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i.e. a lens *internal* to **Cat** between the double categories of squares.

Connection between these characterisations

Lemma

Every split opfibration $(G, k)\colon \mathbf{S} \to \mathbf{V}$ *induces a double category* $\mathbb{B}$ *with domain and codomain maps given by* $\Pi_0, \Pi_1 \colon K \downarrow K \rightrightarrows \Lambda$.

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\end{tikzcd}$$

where $\mathbb{B}$ *is isomorphic to the transpose of* $\mathbb{A} = (G \downarrow \mathbf{V} \rightrightarrows \mathbf{S})$.

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- A delta lens is a split opfibration iff its representation in **Cat** satisfies a condition with respect to the right décalage.
- Every split opfibration may be represented as a diagram in **Dbl** – an *internal lens* between the double categories of squares.
- **Benefits**: Results generalise to **Cat**$(\mathcal{E})$; new proofs of old results on split opfibrations; and clearer understanding.